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Eye-Tracking–Based Measurement of Social Visual Engagement Compared With Expert Clinical Diagnosis of Autism

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IMPORTANCE In the US, children with signs of autism often experience more than 1 year of delay before diagnosis and often experience longer delays if they are from racially, ethnically, or economically disadvantaged backgrounds. Most diagnoses are also received without use of standardized diagnostic instruments. To aid in early autism diagnosis, eye-tracking measurement of social visual engagement has shown potential as a performance-based biomarker.

OBJECTIVE To evaluate the performance of eye-tracking measurement of social visual engagement (index test) relative to expert clinical diagnosis in young children referred to specialty autism clinics.

DESIGN, SETTING, AND PARTICIPANTS In this study of 16- to 30-month-old children enrolled at 6 US specialty centers from April 2018 through May 2019, staff blind to clinical diagnoses used automated devices to measure eye-tracking-based social visual engagement. Expert clinical diagnoses were made using best practice standardized protocols by specialists blind to index test results. This study was completed in a 1-day protocol for each participant.

MAIN OUTCOMES AND MEASURES Primary outcome measures were test sensitivity and specificity relative to expert clinical diagnosis. Secondary outcome measures were test correlations with expert clinical assessments of social disability, verbal ability, and nonverbal cognitive ability.

RESULTS Eye-tracking measurement of social visual engagement was successful in 475 (95.2%) of the 499 enrolled children (mean [SD] age, 24.1 [4.4] months; 38 [8.0%] were Asian; 37 [7.8%], Black; 352 [74.1%], White; 44 [9.3%], other; and 68 [14.3%], Hispanic). By expert clinical diagnosis, 221 children (46.5%) had autism and 254 (53.5%) did not. In all children, measurement of social visual engagement had sensitivity of 71.0% (95% CI, 64.7% to 76.6%) and specificity of 80.7% (95% CI, 75.4% to 85.1%). In the subgroup of 335 children whose autism diagnosis was certain, sensitivity was 78.0% (95% CI, 70.7% to 83.9%) and specificity was 85.4% (95% CI, 79.5% to 89.8%). Eye-tracking test results correlated with expert clinical assessments of individual levels of social disability ($r = -0.75$ [95% CI, -0.79 to -0.71]), verbal ability ($r = 0.65$ [95% CI, 0.59 to 0.70]), and nonverbal cognitive ability ($r = 0.65$ [95% CI, 0.59 to 0.70]).

CONCLUSIONS AND RELEVANCE In 16- to 30-month-old children referred to specialty clinics, eye-tracking–based measurement of social visual engagement was predictive of autism diagnoses by clinical experts. Further evaluation of this test's role in early diagnosis and assessment of autism in routine specialty clinic practice is warranted.

TRIAL REGISTRATION ClinicalTrials.gov Identifier: [NCT03469986](https://clinicaltrials.gov/ct2/show/study/NCT03469986)

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Autism affects approximately 1 in 36 US children.¹ Most parents of children with autism report having had concerns before the second birthday,^{2,3} yet the median age of US autism diagnosis is 4 to 5 years.^{1,4} For US racial and ethnic minority, low-income, and rural families, the age at diagnosis is later still.⁵⁻⁸

Delay in diagnosis delays opportunities to support children and families early, when the brain is most malleable and before maladaptive behaviors may become entrenched.^{9,10} From time of first concern to time of diagnosis, most families face multiple years of referrals and waiting for evaluation at specialty centers.^{11,12}

Objective biomarker tests could offer potential assistance for this issue by aiding in accurate, efficient diagnosis. Measurement of social visual engagement—how children look at and learn from the social environment—offers neurobiological and clinical face validity as a possible measure of autism. Individual variation in social visual engagement reflects individual genetic influence (with monozygotic twin concordance of ~0.9)^{13,14}; developmental change in this factor is evolutionarily conserved¹⁵; and autism-related differences^{13,16,17} emerge at young ages and predict later diagnosis.¹⁶

Previous research developed eye-tracking-based measurement of social visual engagement to aid in early autism diagnosis and assessment.¹⁸ The current study evaluated this test's performance at sites across the US with multiple expert clinical teams and measurements collected on automated eye-tracking devices operated by clinical site staff.

Methods

This study evaluated diagnostic performance of eye-tracking-based measurement of social visual engagement (index test) in comparison with expert clinical diagnosis (reference standard) in terms of both (1) categorical diagnostic labels (autism or nonautism) and (2) continuous measures of social disability, verbal ability, and nonverbal cognitive ability. The trial protocol and statistical analysis plan are in [Supplement 1](#). The study was preregistered and followed Standards for Reporting of Diagnostic Accuracy (STARD) guidelines¹⁹ (see the eMethods in [Supplement 2](#) for details).

Study Design

The study used a multisite, prospective, double-blind, within-subject comparison to evaluate the test's performance among 16- to 30-month-old children enrolled at 6 specialty centers between April 2018 and May 2019, with participants each completing a 1-day protocol. This age range was selected as the period when overt symptoms of autism are commonly observable,²⁰ with the objective for the test to aid in diagnosis prior to 30 months, leaving 6 months in which children could be referred to early intervention before reaching 36 months. Children were enrolled at specialty clinics in the Emory University School of Medicine, Southwest Autism Research & Resource Center, Seattle Children's Autism Center, Cincinnati Children's Hospital, University of California San Francisco, and

Key Points

Question Can eye-tracking-based measurement of social visual engagement aid in early diagnosis and assessment of autism in young children?

Findings In a multisite, prospective, double-blind study of 475 children aged 16 to 30 months assessed for autism in 6 specialty clinics, measurement of social visual engagement had 71.0% sensitivity and 80.7% specificity relative to expert clinical diagnosis. In the subgroup of children whose autism diagnosis was certain ($n = 335$), the test had 78.0% sensitivity and 85.4% specificity.

Meaning Eye-tracking-based measurement warrants further evaluation for early diagnosis and assessment of autism in young children referred to specialty clinics.

Rush University Medical Center. The research protocol was approved as a minimal-risk study by the institutional review boards at each participating institution, and parents or legal guardians gave written informed consent for participating children.

To minimize verification bias,²¹ diagnostic verification was complete (rather than partial or differential). To minimize selection bias,²¹ participants were enrolled consecutively. Study design was double-blind²¹: all staff involved with data collection and analysis of eye tracking were blind to reference standard results, and all staff involved with collection and interpretation of reference standard were blind to eye-tracking results. Only reference standard diagnosis and assessments (current clinical best practice) were used in patient care (neither parents nor study center staff were informed of a child's eye-tracking results).

Two third-party clinical research organizations (STATKING Clinical Services Inc, and Libra Medical Inc) were responsible for study oversight, data management, unblinding, and comparison of index test and reference standard. Although no serious adverse events were anticipated, adverse events were monitored and summarized for safety analyses.

Participants

Children were consecutively enrolled according to the following inclusion criteria: chronological age between 16 and 30 months; normal or corrected-to-normal vision; normal or corrected-to-normal hearing; in generally good health with no acute illness; and parent or guardian able to understand informed consent in English ([Figure 1](#) and [Table](#); eMethods in [Supplement 2](#)). To characterize the study sample, patient demographic data, including race, ethnicity, and maternal education, were collected based on parent report. Race and ethnicity data were collected to enable evaluation of whether test performance varied based on these characteristics.

Reference Standard Diagnosis

Reference standard diagnosis of autism spectrum disorder or nonautism spectrum disorder (labeled herein as autism and nonautism, respectively) was made by expert clinicians using standardized assessments: the Autism Diagnostic Observation

Figure 1. Participant Enrollment and Outcomes Comparing Eye-Tracking-Based Quantification of Social Visual Engagement With Expert Clinical Diagnosis of Autism

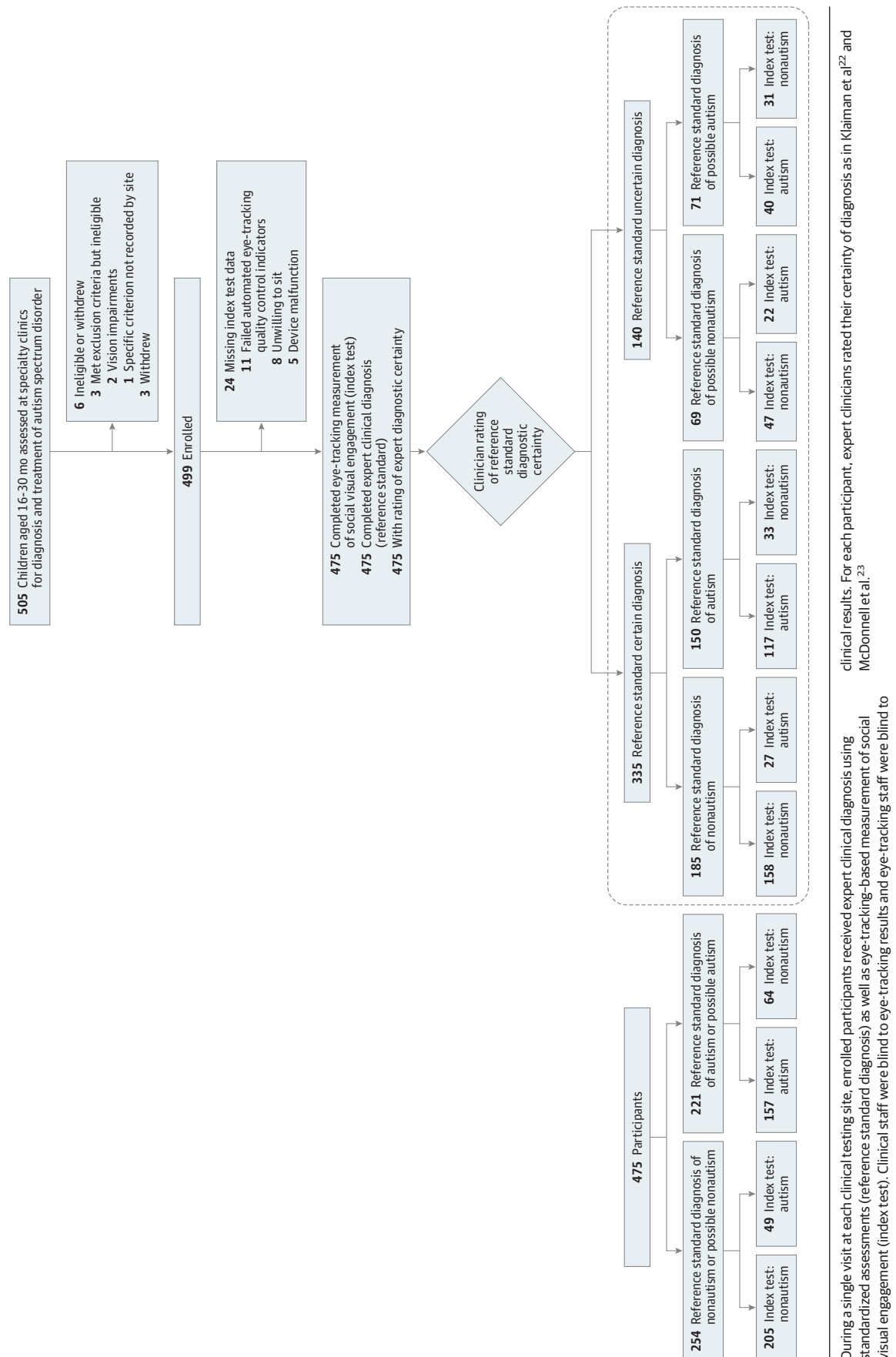


Table. Participant Characterization and Demographics

	Reference standard diagnosis		Certain diagnosis (n = 335)	
	All participants (N = 475) ^a		Autism	Nonautism
No.	Autism or possible autism	Nonautism or possible nonautism	Autism	Nonautism
No.	221	254	150	185
Age, mo				
Mean (SD)	24.9 (4.2)	23.4 (4.5)	24.9 (4.1)	23.4 (4.6)
Median (IQR)	26 (21-29)	24 (19-27)	26 (22-29)	24 (19-28)
Sex, No. (%)				
Female	57 (25.8)	127 (50.0)	38 (25.3)	102 (55.1)
Male	164 (74.2)	127 (50.0)	112 (74.7)	83 (44.9)
Race, No. (%) ^b				
Asian	21 (9.5)	17 (6.7)	14 (9.3)	10 (5.4)
Black/African American	22 (9.9)	15 (5.9)	16 (10.7)	11 (5.9)
Native Hawaiian or Other Pacific Islander	2 (0.9)	0	1 (0.7)	0
White	149 (67.4)	203 (79.9)	95 (63.3)	147 (79.5)
Other	26 (11.8)	18 (7.1)	23 (15.3)	17 (9.2)
Unknown	1 (0.5)	1 (0.4)	1 (0.7)	0
Ethnicity, No. (%) ^c				
Hispanic	44 (19.9)	24 (9.4)	26 (17.3)	16 (8.7)
Non-Hispanic	177 (80.1)	225 (88.6)	124 (82.7)	166 (89.7)
No response	0	5 (2.0)	0	3 (1.6)
Maternal education, No. (%) ^d				
Less than eighth grade	0	1 (0.4)	0	1 (0.5)
Some high school	4 (1.8)	3 (1.2)	3 (2.0)	3 (1.6)
High school or GED	33 (14.9)	12 (4.7)	23 (15.3)	6 (3.2)
Some college, no degree	39 (17.7)	22 (8.6)	31 (20.7)	12 (6.5)
Vocational school	12 (5.4)	2 (0.8)	7 (4.7)	0
Associate's degree	13 (5.9)	13 (5.1)	8 (5.4)	12 (6.5)
Bachelor's degree	71 (32.1)	79 (31.1)	50 (33.3)	59 (31.9)
Master's degree	41 (18.6)	82 (32.3)	23 (15.3)	59 (31.9)
Professional or doctoral degree	2 (0.9)	35 (13.8)	0	29 (15.7)
No response	6 (2.7)	5 (2.0)	5 (3.3)	4 (2.2)
ADOS-2 ^{e,f}				
Social affect score, mean (SD)	14.7 (4.3)	2.9 (3.0)	16.1 (3.5)	2.1 (2.1)
Median (IQR)	16 (11-18)	2 (1-4)	17 (14-19)	1 (0-3)
RRB score, mean (SD)	4.9 (2.0)	1.2 (1.3)	5.4 (1.9)	0.9 (1.0)
Median (IQR)	5 (4-6)	1 (0-2)	6 (4-7)	1 (0-1)
Total score, mean (SD)	19.6 (5.1)	4.1 (3.5)	21.4 (4.1)	2.9 (2.4)
Median (IQR)	20 (16-24)	3 (2-6)	22 (18-24)	3 (1-4)

(continued)

Table. Participant Characterization and Demographics (continued)

	Reference standard diagnosis		Certain diagnosis (n = 335)	
	All participants (N = 475) ^a	Autism or possible autism	Nonautism or possible nonautism	Autism
Mullen Scales of Early Learning ^b				
Verbal age equivalent score, mean (SD), mo	12.5 (7.1)	24.5 (8.3)	10.5 (5.3)	26.0 (8.2)
Median (IQR)	11 (8-16)	24 (19-31)	9 (7-13)	25 (20-32)
Nonverbal age equivalent score, mean (SD), mo	18.5 (5.9)	26.5 (8.7)	17.3 (5.0)	27.4 (9.0)
Median (IQR)	18 (15-21)	25 (20-30)	17 (14-21)	26 (21-31)
Other nonautism diagnoses ^c				
Developmental disability	124 (56.1)	210 (82.7)	86 (57.3)	162 (87.6)
No other diagnoses (unaffected)	0	44 (17.3)	0	23 (12.4)

Abbreviations: ADOS-2, Autism Diagnostic Observation Schedule, second edition; RRB, restricted and repetitive behavior.

^a For a table summarizing participant characterization and demographics data for all participants eligible and enrolled (N = 499), irrespective of diagnostic outcomes, please see eTable 1 in Supplement 3.

^b Race data were collected as fixed categories by parental selection, with both "other" and "unknown" as options.

^c Ethnicity data were collected as fixed categories by parental selection.

^d Maternal education data were collected as fixed categories by parental selection.

^e ADOS-2 is a standardized diagnostic assessment for autism, administered by a trained clinical specialist using a semistructured play session consisting of a set of social and communication interactions intended to elicit behaviors relevant to autism diagnosis. The social affect domain score includes test items pertaining to communication and reciprocal social interaction; higher scores indicate more autism symptoms: scores shift slightly based on module and age of child, with scores of approximately 0 to 6 indicating minimal symptoms, scores of approximately 7 to 10 indicating moderate symptoms, and approximately 11 and greater indicating more significant symptoms. Score range is from 0 to 20 (toddler module for children with few to no words) or 0 to 22 (module 2 and toddler module for older children with some words). RRB domain score, which includes test items pertaining to restricted and repetitive behaviors; higher scores indicate more autism symptoms: scores of approximately 1 to 2 indicate minimal symptoms, scores of approximately 3 to 4 indicate moderate symptoms, and approximately 5 and greater indicate more significant symptoms. Score range is from 0 to 6 (toddler module for children with few to no words) or 0 to 8 (module 2 and toddler module for older children with some words). Total score is the total of social affect and RRB. Higher scores indicate more autism symptoms, with

scores shifting slightly based on module and age of child: scores from approximately 0 to 7/9 indicate minimal concern, from approximately 8 to 11 or 10 to 13 indicate mild to moderate concern, and approximately 12/14 and greater indicate more moderate to severe concern. Ranges vary according to the toddler module algorithm (children with few to no words or older children with some words) or module 2. In this sample, 414 (87.5%) children received the toddler module.

^f There were 252 children with nonautism diagnoses (2 children missing data because they were not yet walking, no ADOS-2 conducted because administration would be considered invalid).

^g The Mullen Scales of Early Learning is a standardized developmental assessment, administered by a trained clinical specialist, to measure a child's language, motor, perceptual, and cognitive abilities. Mullen verbal ability age equivalent score, in months, calculated as the average of the Mullen expressive language age equivalent score and the Mullen receptive language age equivalent score. Scores range from 1 to 70 months, with higher scores indicating more advanced receptive and expressive language abilities. Mullen nonverbal ability age equivalent score, in months, calculated as Mullen visual reception age equivalent score. Scores range from 1 to 69 months, with higher scores indicating more advanced nonverbal problem-solving abilities. Mullen age equivalents indicate the age bracket in the Mullen norming sample that had the same median score as a tested child's raw score. Age equivalents are helpful in showing differences between performance areas or in emphasizing that a child's score was similar to or different from abilities expected at a given chronological age.

^h Expert clinician diagnosis of 1 or more co-occurring developmental disabilities, including language, cognitive, or motor delays, vs no clinical diagnosis of any kind. Speech-language and global developmental delays were the most common other diagnoses given.

Schedule, Second Edition (ADOS-2)²⁴; the Mullen Scales of Early Learning (Mullen)²⁵; medical and developmental histories; and DSM-5 criteria. Expert autism clinician status was confirmed on the basis of clinician training and experience and by establishing ADOS-2 research reliability with a certified ADOS-2 trainer (eMethods in [Supplement 2](#)).

These procedures constitute current best practice for diagnosing autism, but are nonetheless based on the expert clinician's subjective judgment. Given this, together with the study's focus on diagnosing very young children^{20,26} who can present with complex, evolving clinical profiles (eg, language, cognitive, or other challenges, co-occurring with or distinct from autism), expert clinicians were required to prospectively rate their level of certainty in each diagnosis.^{22,23} Reference standard certainty is important because performance metrics of comparison tests depend inherently on the reliability and validity of the reference standard: if a participant's reference standard diagnosis is uncertain, then that participant's ground truth for comparison will also be uncertain. Certainty was rated prospectively with each expert clinician blinded to index test results.^{22,23}

Index Test Measurement of Social Visual Engagement

Social visual engagement data were collected using 6 investigational eye-tracking devices, one at each clinical site. Device operators were staff technicians employed by each clinical site, trained for approximately 1 hour when each device was delivered, with no other required specialization or prior training. The device was located in available space within the clinic (ie, examination room, playroom). Data collection procedures were automated, with no instructions required of the child. Software guided the device operator through eye-tracking calibration, validation of calibration, and automated data collection.

The eye-tracking index test measured social visual engagement while children watched video scenes of social interaction, as previously described^{13,18} (see also the Experimental Procedures section under eMethods in [Supplement 2](#)). As in the study by Jones et al,¹⁸ 14 video scenes were presented, each with a mean (SD) duration of about 54.0 (21.5) seconds (range, 21.7 seconds to 1 minute 29.7 seconds).

Following collection, index test data were automatically uploaded to a secure cloud-based server for processing and analysis. Processing and analysis were automated, and each participant's data, once analyzed, yielded categorical determination of either autism or nonautism, as well as 3 continuous measures: a social disability index, verbal ability index, and nonverbal ability index (see the Data Processing and Data Analysis sections under eMethods in [Supplement 2](#) for details).

Planned Analyses

Primary outcome analyses compared results from the eye-tracking index test with results from reference standard diagnosis. The prespecified test positivity threshold was determined in the efficacy study,¹⁸ fixed, and applied herein (values ≤ 0 indicated autism). The comparison was quantified as index test sensitivity and specificity, with 2-sided 95% CIs,²⁷ to-

gether with receiver operating characteristic curves and area under the curve metrics. Positive predictive value (PPV), negative predictive value (NPV), and accuracy of the index test were also calculated. Primary end point analyses were tested at a 1-sided significance level of $\alpha = .025$.

Secondary outcome analyses compared eye-tracking-based indices of social disability, verbal ability, and nonverbal cognitive ability with their respective expert clinician-administered reference standard assessments (ADOS-2 total scores, Mullen verbal age equivalent scores [average of receptive and expressive language], and Mullen nonverbal age equivalent scores). Standard regression diagnostics were performed for all comparisons,^{28,29} and Deming regression^{30,31} was used to estimate regression coefficients. Pearson correlation coefficients and adjusted R^2 coefficients of determination,³²⁻³⁴ together with 95% CIs, were calculated. Secondary outcome analyses were tested at a 1-sided significance level of $\alpha = .025$.

Exploratory Analyses

Exploratory analyses measured effects of race, ethnicity, sex, age, eye-tracking calibration accuracy, false-positive and false-negative clinical outcomes, reference standard certainty, and reference standard measurement error. Reference standard certainty sets an upper limit on performance metrics of comparison tests, but also enabled a post hoc inferential hypothesis test: if an index test is accurate, its performance should be (1) superior when the reference standard is certain and (2) inferior when the reference standard is uncertain (an accurate test should not match an uncertain reference standard). Similarly, reference standard measurement error variance sets an upper limit on the amount of nonerror variance that can be explained by comparison tests³²⁻³⁴; reference standard instruments in the current study have "good" but not "excellent" test-retest reliability^{24,25} (0.87, 0.74, and 0.75, respectively, for ADOS-2 total score and Mullen verbal and nonverbal scales). Finally, we compared the present study results with findings from prior studies¹⁸ to assess replicability.

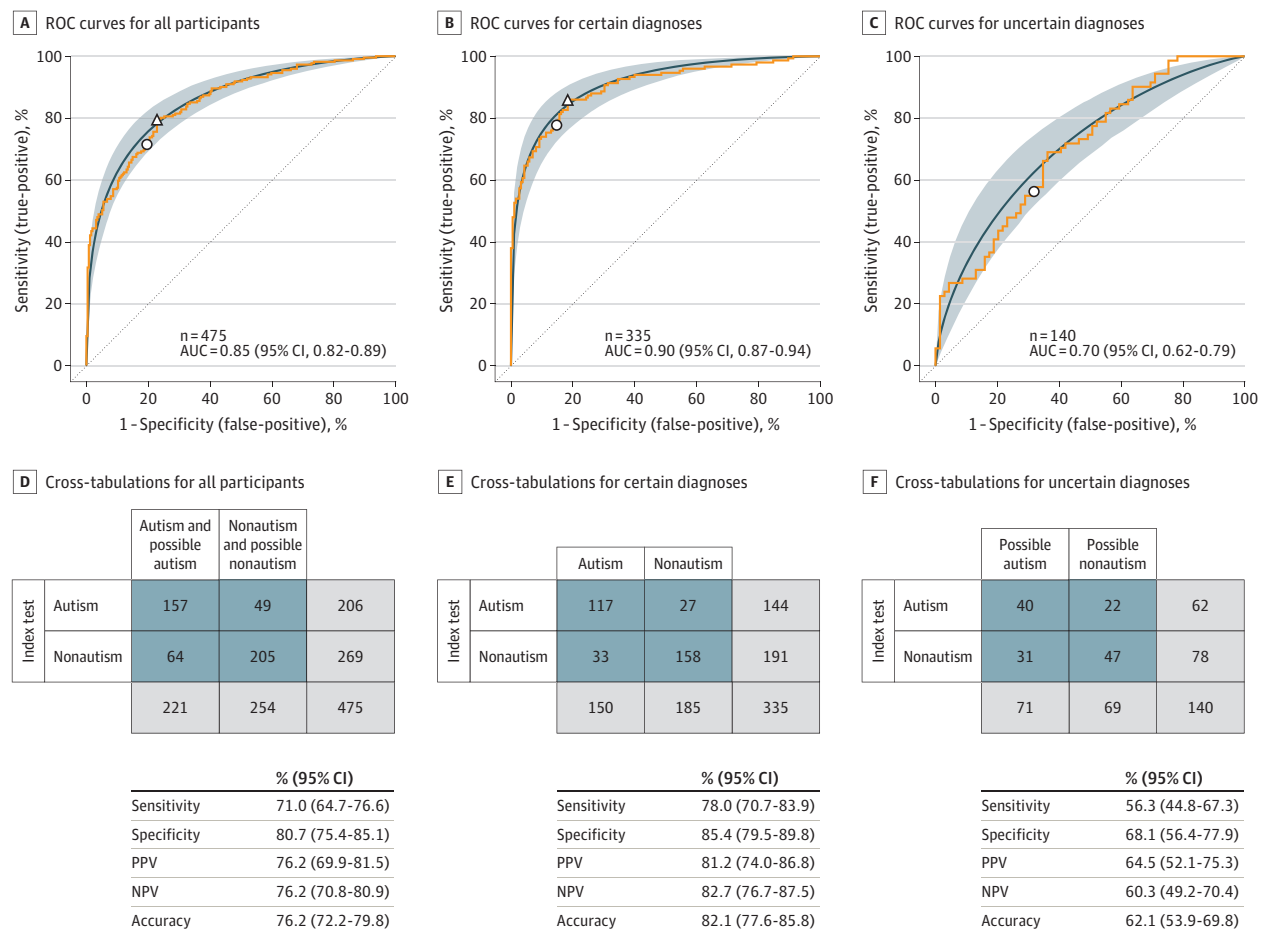
Results

Participants

Of 499 eligible children enrolled, 475 (95.2%) successfully completed the eye-tracking test and comprise the primary analysis set (Figure 1). Expert clinicians were certain of reference standard diagnoses in 335 (70.5%) of the 475 participants, while there was uncertainty in 140 (29.5%).²²

The Table describes participant demographic and clinical characteristics. The mean (SD) age of participants was 24.1 (4.4) months. Additionally, 38 (8.0%) were Asian; 37 (7.8%), Black; 352 (74.1%), White; 44 (9.3%), other; and 68 (14.3%) were of Hispanic ethnicity. Participants with autism had higher ADOS-2 domain and total scores (all $t > 23.8$, all $P < .001$). They also had lower Mullen verbal age equivalent scores and lower Mullen nonverbal age equivalent scores ($t = 17.0$, $P < .001$ and $t = 11.8$, $P < .001$, respectively). Characteristics of all eligible enrolled participants, including those who did not complete the test, are provided in eTable 1 in [Supplement 3](#), and comparison of

Figure 2. Test Performance of Measurement of Social Visual Engagement (Index Test) vs Reference Standard Diagnosis of Autism



Receiver operating characteristic (ROC) curves for comparison of index test performance relative to reference standard diagnosis in all participants (A), participants for whom expert clinicians rated their diagnoses as certain (B), and participants for whom expert clinicians rated their diagnoses as uncertain (C). Empirical area under the curve (AUC) metrics and 95% CIs are reported on each ROC plot. In panels A, B, and C, the prespecified test positivity threshold is marked with circles. The prespecified test positivity threshold was determined in the efficacy study,¹⁸ fixed, and applied here in the pivotal multisite trial. For comparison, a theoretical optimal threshold determined post hoc by Youden

Index is marked with triangles in panels A and B. In panels D, E, and F, cross-tabulations of the achieved eye-tracking index test results vs reference standard diagnoses are given together with corresponding test performance estimates and 95% CIs. Performance results in D, E, and F all correspond to the circles marked in A, B, and C. Negative predictive value (NPV) and positive predictive value (PPV) estimates reported here are calculated based on study sample prevalence. In panels A, B, and C, the orange lines indicate the empirical ROC curve, while blue lines and shaded areas indicate the fitted ROC estimates with 95% CIs, respectively.

certain and uncertain diagnosis subsamples in eTable 2 in Supplement 3. Eye-tracking quality control indicators are provided in eFigure 1 and eTable 3 in Supplement 3.

Among nonautism and possible nonautism participants, 82.7% had 1 or more co-occurring cognitive or developmental delays (210/254, based on Mullen scores), while 17.3% (44/254) did not. Likewise, among autism and possible autism participants, 56.1% had co-occurring cognitive or developmental delays (124/221), while 43.9% did not (97/221).

While chronological age was restricted to 16 to 30 months, children's demonstrated verbal abilities (in both diagnostic groups) varied from those delayed to levels approximating typical 4-month-old infants (eg, vocalizing with nonspecific syllable sounds³⁵) to those with advanced skills more common to 44-month-old children (3.7 years old) (ie, knowing several hundred or more words; producing competent phrase

speech^{25,35}). For nonverbal abilities, the range varied similarly from skills approximating those of typical 8-month-old infants (eg, working on fine grasping; beginning to actively seek out-of-view objects³⁵) to those more common to typical 50-month-old children (4.2 years old) (independently performing self-help skills; knowing multiple numbers and amounts³⁵).

Primary End Points: Estimates of Diagnostic Accuracy

Figure 2 presents results for the primary effectiveness end point: accuracy of diagnostic classification when compared with reference standard expert clinical diagnosis (see eFigure 2 in Supplement 3 for underlying distribution data). Figure 2A plots the receiver operating characteristic curve for all participants with both index test and reference standard diagnosis (primary analysis set). As noted above, this includes

data for 140 participants (29.5% of the sample) for whom clinicians were uncertain of their reference standard diagnosis; despite that uncertainty, index test performance had area under the curve equal to 0.85 (95% CI, 0.82-0.89), sensitivity of 71.0% (95% CI, 64.7%-76.6%), and specificity of 80.7% (95% CI, 75.4%-85.1%). Crosstabulation data and performance metrics are given in Figure 2D. These results represent worst-case performance given the amount of reference standard uncertainty in the comparison (29.5%) because reference standard uncertainty necessarily sets an upper limit on comparison test metrics (eFigure 3 in Supplement 3).

For the 335 children for whom the reference standard was certain (Figure 2B), index test sensitivity was 78% (95% CI, 70.7%-83.9%) and specificity was 85.4% (95% CI, 79.5%-89.8%), with area under the curve of 0.90 (95% CI, 0.87-0.94). Accuracy relative to the reference standard was 82.1% (95% CI, 77.6%-85.8%). Crosstabulation data and performance metrics are given in Figure 2E.

In contrast, for the 140 children for whom the reference standard was uncertain, ie, when the expert clinicians judged their own diagnoses as uncertain, the index test's sensitivity was 56.3% (95% CI, 44.8%-67.3%) and specificity was 68.1% (95% CI, 56.4%-77.9%) (Figure 2C and F). The index test's performance metrics were significantly better when the reference standard was certain relative to uncertain ($z = 4.65$, $P < .001$).

The present study results replicated prior findings from laboratory-based settings¹⁸ (eFigures 4 and 5 in Supplement 3).

In exploratory analyses connecting results in Figure 2 to participant demographics in the Table, reference standard diagnoses in this sample were not statistically independent of racial categories ($\chi^2_5 = 11.24$, $P = .048$): children whose parents selected racial categories of either Black/African American or other were given significantly more autism diagnoses, while children whose parents selected White were given significantly fewer autism diagnoses. Likewise, children identified by their parents as Hispanic or Latino were given proportionally more autism diagnoses, while children identified as non-Hispanic or Latino were given proportionally fewer autism diagnoses ($\chi^2_2 = 14.39$, $P < .001$). In contrast, when these tests were instead conducted by grouping children by their eye-tracking results, no significant associations were observed (eTable 4 in Supplement 3). Thus, performance-based eye-tracking results were not associated with race and ethnicity, while reference standard results were (see the Associations With Racial and Ethnic Category section under eResults in Supplement 3 for details).

Secondary Outcomes: Measurements of Symptom Severity

For social disability, correlation between the index test social disability index and the ADOS-2 total score was -0.75 (95% CI, -0.79 to -0.71) (Figure 3A). For verbal ability, correlation with the Mullen verbal age equivalent score was 0.65 (95% CI, 0.59 to 0.70) (Figure 3B); and for nonverbal ability, correlation with the Mullen nonverbal age equivalent score was 0.65 (95% CI, 0.59 to 0.70) (Figure 3C). Score comparisons between index test and reference standards are given in eTables 5, 6, and 7 in Supplement 3).

Exploratory analyses were conducted to evaluate the percentage of variance in the reference standards that could be explained by eye-tracking test results while accounting for test-retest reliability of the reference standards (eTable 8 in Supplement 3).³²⁻³⁴ The eye-tracking-based Social Disability Index accounted for 74.1% of variance in ADOS-2 total scores (adjusted $R^2 = 0.74$ [95% CI, 0.66 - 0.81]). The eye-tracking-based Verbal Ability Index accounted for 75.1% of variance in Mullen verbal age equivalent scores (adjusted $R^2 = 0.75$ [95% CI, 0.66 - 0.88]). The eye-tracking-based Nonverbal Ability Index accounted for 74.2% of variance in Mullen nonverbal age equivalent scores (adjusted $R^2 = 0.74$ [95% CI, 0.63 - 0.85]). Observed associations did not differ significantly between children whose autism diagnosis was certain compared with uncertain (eTable 8 in Supplement 3).

Again, the present study results replicated prior findings¹⁸ (eFigures 4 and 5 in Supplement 3).

Individual Positive and Negative Predictive Values

Figure 4A presents a flowchart of positive predictive value (PPV) and negative predictive value (NPV) for categorical results within the study. Underlying the categorical results, each child has a continuous numerical score that is information-bearing for diagnostic probability (see eFigure 2 in Supplement 3 for density distributions of all numerical scores). Figure 4B presents continuous PPV and NPV curves as a function of the underlying numerical score, together with scores for 4 example children from the study. Scores at either extreme of the PPV and NPV curves substantially increase predictive probability of autism or nonautism diagnoses, while intermediate scores present intermediate probabilities. Figure 4C summarizes the pretest, categorical post-test, and individual post-test probabilities of diagnoses for each example child. PPV and NPV are important because they quantify the real-world choice faced by clinicians: with a positive or negative test result and associated underlying biomarker score, what is the probability that this child does or does not have the condition?

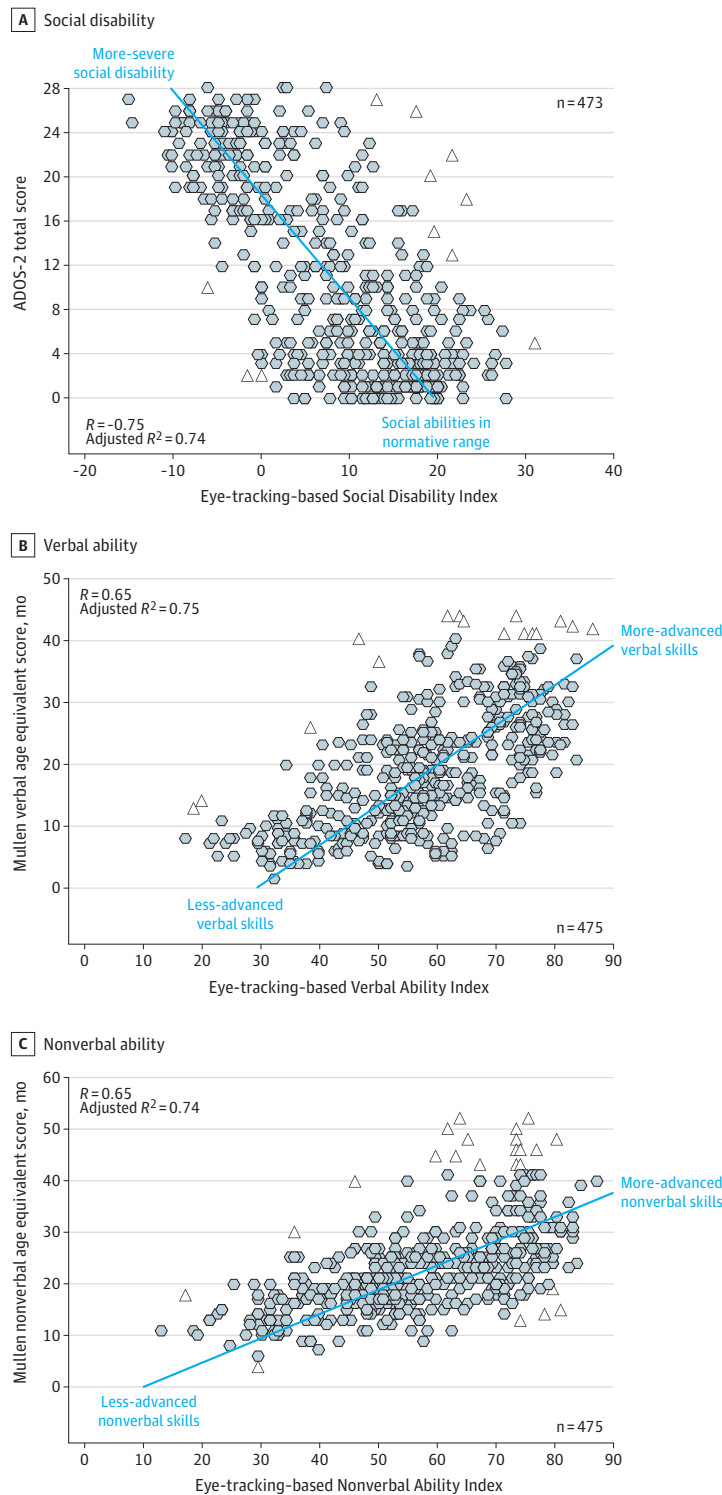
The values in Figure 4 are calculated based on current study sample prevalence, with prior probability of autism diagnosis equal to 44.8% (this is, of course, substantially greater than general population prevalence due to referrals for concerns). It should be noted that PPV and NPV of any test vary mathematically with prevalence.³⁶ eFigure 6 in Supplement 3 presents that variation as well as information regarding PPV and NPV in different clinical contexts.

For additional information, see eResults in Supplement 3.

Discussion

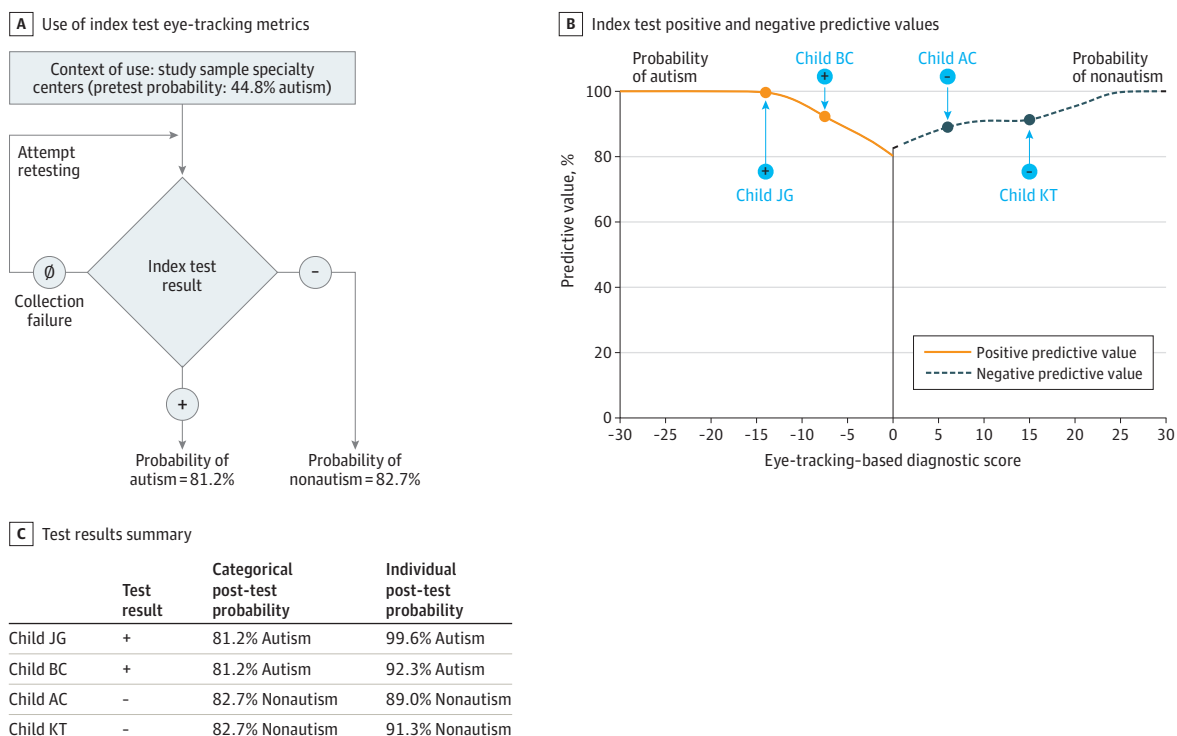
This study found that eye-tracking-based measurement of social visual engagement had sensitivity of 71.0% and specificity of 80.7% to predict expert clinical diagnosis of autism in children aged 16 to 30 months seen at 6 specialty centers. Sensitivity was 78.0% and specificity was 85.4% when expert clinicians were certain of their diagnoses. Strengths of the present study include its multisite, prospective, double-blind design and its evaluation of specific autism-related outcomes including social disability and verbal and nonverbal ability.

Figure 3. Correlation Between Eye-Tracking-Based Measurement of Social Visual Engagement and Clinician-Administered, Standardized Assessments of Social Disability, Verbal Ability, and Nonverbal Cognitive Ability



A, Correlation between eye-tracking-based indices of social disability vs children's total scores on the Autism Diagnostic Observation Schedule, Second Edition (ADOS-2, a standardized diagnostic assessment for autism, administered by a trained clinical specialist using a semistructured play session consisting of a set of presses for social and communication interaction intended to elicit behaviors relevant to autism diagnosis). B, Correlation between eye-tracking-based indices of verbal ability vs children's verbal age equivalent scores as measured by the Mullen Scales of Early Learning (standardized developmental assessment, administered by a trained clinical specialist, to measure a child's language, motor, perceptual, and cognitive abilities). C, Correlation between eye-tracking-based indices of nonverbal cognitive ability vs children's nonverbal age equivalent scores as measured by the Mullen Scales of Early Learning. The blue lines in each panel are Deming regression fitted functions, with labels at each end that give directional interpretation for each measurement comparison. In all scatter plots, the blue data markers indicate individual data, and the triangles indicate regression outliers (bivariate outliers identified using Cook distance and difference-in-fits regression diagnostics). Adjusted R^2 values are adjusted for test-retest reliability of the reference standard (yielding percentage of reference standard nonerror variance explained by the index test). See the eMethods in Supplement 2 (Secondary Endpoint Analyses) for additional information.

Figure 4. Index Test Predictive Value for Individual Children



A, Flowchart for utilization of index test eye-tracking metrics in specialty centers serving as referral hubs for children with developmental concerns (shown here as having prior probability of autism equal to observed study sample prevalence, 44.8%, for children with certain reference standard; referral center prevalence was expected to be higher than general prevalence due to referrals for concerns). Categorical test results provide post-test probabilities for either autism or nonautism diagnoses. B, Index test positive predictive values (orange line) and negative predictive values (dashed blue-gray line) vary continuously with diagnostic score. Example individual results for 4 children (JG, BC, AC, and KT; drawn from current study sample) are marked with arrows

and circles, indicating the test score and the score's corresponding positive or negative predictive value. C, Summary of test results, categorical post-test predictive value, and individual post-test predictive value for the 4 individual children plotted in panel B. Scores toward either end of the continuum increase probability of autism or nonautism diagnoses, while intermediate scores present intermediate probabilities. All positive and negative predictive values are adjusted for expected prevalence in the context of use. For additional examples of how positive and negative predictive values vary with prevalence, see eFigure 6 in Supplement 3.

The test performance observed in this study should be evaluated in the context of this being a diagnostically complex sample. It included a very broad spectrum of children with autism—both with and without co-occurring intellectual and developmental delays—as well as many children without autism who nonetheless had significant speech-language and global developmental disabilities. The challenge of reference standard differential diagnosis in this sample was highlighted by clinician uncertainty in 29.5% of children, which is comparable with, yet slightly lower than, other reports in the literature (of up to 40%²³).

The tool used in this study resulted in successful test collection in 95% of the children enrolled. Data were collected on automated eye-tracking devices, operated by staff technicians with approximately 1 hour of training in 6 different specialty clinics. The test is objective, performance-based, quantitative, and directly measures thousands of instances of child behavior for comparison with age-expected norms.¹⁸ The data collection burden for the child is low in that it involves watching videos, and results can be generated in less than 30 minutes.

Results of standardized assessments were predicted by measurements of social visual engagement, with index test

measurements explaining 74.1% of variance in children's individual levels of social disability, 75.1% of variance in verbal ability, and 74.2% of variance in nonverbal cognitive ability, all as assessed by best practice reference standard instruments. Administration of these instruments otherwise requires multiple hours of testing per child as well as extensive training and expertise. In comparison, processing and analysis of social visual engagement data, although computationally intensive,¹⁸ is entirely automated and cloud-based, capable of returning results to the clinician—including generating each child's report—within half an hour.

More broadly, key advances in the identification, prognosis, and monitoring of medical conditions have often started with discovery and validation of objective biomarkers.³⁷ The present results suggest that measurement of social visual engagement—how children look at and learn from the surrounding social environment—may offer such a biomarker for autism. Ideally, utilization of such a biomarker has potential to shorten the time from first concern to diagnosis, and thereby accelerate time to intervention when appropriate. It also has potential to provide quantitative endophenotypes for research in genetics and neurobiology, and may have potential

to stratify subgroups and monitor treatment outcomes in clinical trials.

Measurement of social visual engagement is a transactional biomarker. Methodologically, this approach leverages the environmental context—the social content presented to patients—to amplify relevant patient signal and thus increase effect size for measurable biomarker detection (akin to a treadmill test in cardiology). Conceptually, this approach recognizes that disability, if it exists, is a consequence of individual vulnerabilities transacting with particular contexts, severely disabling in some but less so or not at all in others.³⁸

This transactional principle is also consistent with what happens in expert clinical diagnosis, when a clinician assesses a child and presents repeated, semistructured social overtures to elicit and gauge a child's response. With eye-tracking measurements of social visual engagement, the principle is similar but the difference is in the extent of quantification and in number of measurements collected. The technology may be used to extend a clinician's reach by quickly presenting and quantifying thousands of moment-by-moment transactional reactions from each unique child.¹⁸ The end goals are the same: to enable earlier and more efficient diagnosis; to monitor behaviors that may exacerbate or ameliorate

disability; and to ultimately personalize interventions on the basis of individual needs.

Limitations

A limitation of this study was that it was not intended to test general population screening. Studies in general population settings, where the prevalence of autism is lower than in the specialty clinics in the current study, will be of interest. Relatedly, the tool evaluated in this study is not intended to replace expert clinicians, but to supplement informed and experienced clinical judgment. Autism is a consequential diagnosis, and families of children with autism benefit from clinicians' support to translate assessment results into plans of care and support as needed.

Conclusions

In 16- to 30-month-old children referred to specialty clinics, measurement of social visual engagement was predictive of autism diagnoses by clinical experts. Further evaluation of this test's role in early diagnosis and assessment of autism in routine specialty clinic practice is warranted.

ARTICLE INFORMATION

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Conflict of Interest Disclosures: Dr Jones reported being employed by Emory University and Marcus Autism Center, a subsidiary of Children's Healthcare

of Atlanta (CHOA); and receiving grants from The Marcus Foundation, the Joseph B. Whitehead Foundation, and the Georgia Research Alliance. Marcus Autism Center was the study sponsor during trial data collection. The technology and study reporting responsibilities were transferred to EarliTec Diagnostics Inc as part of technology transfer from CHOA on January 31, 2020. Dr Jones was a scientific co-founder of EarliTec and is an equity holder, and acts as a paid scientific consultant to EarliTec and unpaid board member, although his primary appointment and affiliation remain at Emory. EarliTec consulting fees are paid to Dr Jones. Dr Jones also reported a lecture honorarium from Washington University School of Medicine in St Louis (Third Annual Dr Adolfo and Fanny Rizzo Endowed Lecture). In addition, Dr Jones reported US patent No.s 7,922,670; 8,343,067; 8,551,015; 9,265,416; 9,510,752; 9,861,307; 10,016,156; 10,022,049; 10,617,295; 10,702,150; and 10,987,043, licensed to EarliTec Diagnostics Inc. Dr Klaiman reported being employed by Emory University and Marcus Autism Center, a subsidiary of CHOA; and receiving grants from The Marcus Foundation, the Joseph B. Whitehead Foundation, and the Georgia Research Alliance. Dr Klaiman acts as a paid scientific consultant to EarliTec Diagnostics Inc and as a clinical advisor to Beaming Health, and reported personal fees from EarliTec and Beaming Health. Dr Klaiman is a certified ADOS-2 trainer and received personal fees to conduct ADOS-2 trainings at ABA Centers of America, Dekalb County School District, Cherokee County School District, and Fulton County School District. Dr Klaiman also reported an honorarium from Children's Health Council. Dr Richardson reported being employed by Emory University and Marcus Autism Center, a subsidiary of CHOA; and receiving funds and/or equipment to the institution from The Marcus Foundation, the Joseph B. Whitehead Foundation, and the Georgia Research Alliance. Dr Aoki reported

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scientific consultant to EarliTec and as an unpaid board member, although his primary appointment and affiliation remain at Emory. EarliTec consulting fees are paid to Dr Klin. Dr Klin also reported lecture honoraria from National Autism Conference, McKnight Endowment Fund for Neuroscience, Sociedade Brasileira de Fonoaudiologia, Alliance for Early Success, and Washington University School of Medicine; in addition, Dr Klin reported US patent Nos 7,922,670; 8,343,067; 8,551,015; 9,265,416; 9,510,752; 9,861,307; 10,016,156; 10,022,049; 10,617,295; 10,702,150; and 10,987,043 licensed to EarliTec Diagnostics Inc. No other disclosures were reported.

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REFERENCES

- Maenner MJ, Warren Z, Williams AR, et al. Prevalence and characteristics of autism spectrum disorder among children aged 8 years: Autism and Developmental Disabilities Monitoring Network, 11 sites, United States, 2020. *MMWR Surveill Summ*. 2023;72(2):1-14. doi:10.15585/mmwr.ss7202a1
- Chawarska K, Paul R, Klin A, Hannigen S, Dichtel LE, Volkmar F. Parental recognition of developmental problems in toddlers with autism spectrum disorders. *J Autism Dev Disord*. 2007;37(1):62-72. doi:10.1007/s10803-006-0330-8
- Wetherby AM, Brosnan-Maddox S, Peace V, Newton L. Validation of the Infant-Toddler Checklist as a broadband screener for autism spectrum disorders from 9 to 24 months of age. *Autism*. 2008;12(5):487-511. doi:10.1177/1362361308094501
- Shaw KA, McArthur D, Hughes MM, et al. Progress and disparities in early identification of autism spectrum disorder: Autism and Developmental Disabilities Monitoring Network, 2002-2016. *J Am Acad Child Adolesc Psychiatry*. 2022;61(7):905-914. doi:10.1016/j.jaac.2021.11.019
- Shattuck PT, Durkin M, Maenner M, et al. Timing of identification among children with an autism spectrum disorder. *J Am Acad Child Adolesc Psychiatry*. 2009;48(5):474-483. doi:10.1097/CHI.0b013e31819b3848
- Mandell DS, Wiggins LD, Carpenter LA, et al. Racial/ethnic disparities in the identification of children with autism spectrum disorders. *Am J Public Health*. 2009;99(3):493-498. doi:10.2105/AJPH.2007.131243
- Constantino JN, Abbacchi AM, Saulnier C, et al. Timing of the diagnosis of autism in African American children. *Pediatrics*. 2020;146(3):e20193629. doi:10.1542/peds.2019-3629
- Fountain C, King MD, Bearman PS. Age of diagnosis for autism. *J Epidemiol Community Health*. 2011;65(6):503-510. doi:10.1136/jech.2009.104588
- Dawson G, Jones EJH, Merkle K, et al. Early behavioral intervention is associated with normalized brain activity in young children with autism. *J Am Acad Child Adolesc Psychiatry*. 2012;51(11):1150-1159. doi:10.1016/j.jaac.2012.08.018
- Shultz S, Klin A, Jones W. Neonatal transitions in social behavior and their implications for autism. *Trends Cogn Sci*. 2018;22(5):452-469. doi:10.1016/j.tics.2018.02.012
- Lappé M, Lau L, Dudovitz RN, Nelson BB, Karp EA, Kuo AA. The diagnostic odyssey of autism spectrum disorder. *Pediatrics*. 2018;141(suppl 4):S272-S279. doi:10.1542/peds.2016-4300C
- Kanne SM, Bishop SL. Editorial perspective: the autism waitlist crisis and remembering what families need. *J Child Psychol Psychiatry*. 2021;62(2):140-142. doi:10.1111/jcpp.13254
- Constantino JN, Kennon-McGill S, Weichselbaum C, et al. Infant viewing of social scenes is under genetic control and is atypical in autism. *Nature*. 2017;547(7663):340-344. doi:10.1038/nature22999
- Kennedy DP, D'Onofrio BM, Quinn PD, Bölte S, Lichtenstein P, Falck-Ytter T. Genetic influence on eye movements to complex scenes at short timescales. *Curr Biol*. 2017;27(22):3554-3560.e3. doi:10.1016/j.cub.2017.10.007
- Wang A, Payne C, Moss S, Jones WR, Bachevalier J. Early developmental changes in visual social engagement in infant rhesus monkeys. *Dev Cogn Neurosci*. 2020;43:100778. doi:10.1016/j.dcn.2020.100778
- Jones W, Klin A. Attention to eyes is present but in decline in 2-6-month-old infants later diagnosed with autism. *Nature*. 2013;504(7480):427-431. doi:10.1038/nature12715
- Klin A, Lin DJ, Gorrindo P, Ramsay G, Jones W. Two-year-olds with autism orient to non-social contingencies rather than biological motion. *Nature*. 2009;459(7244):257-261. doi:10.1038/nature07868
- Jones W, Klin A, Richardson S, et al. Development and replication of objective measurements of social visual engagement to aid in early diagnosis and assessment of autism. *JAMA Netw Open*. Published September 5, 2023. doi:10.1001/jamanetworkopen.2023.30145
- Cohen JF, Korevaar DA, Altman DG, et al. STARD 2015 guidelines for reporting diagnostic accuracy studies. *BMJ Open*. 2016;6(11):e012799. doi:10.1136/bmjopen-2016-012799
- Hyman SL, Levy SE, Myers SM; Council on Children With Disabilities, Section on Developmental and Behavioral Pediatrics. Identification, evaluation, and management of children with autism spectrum disorder. *Pediatrics*. 2020;145(1):e20193447. doi:10.1542/peds.2019-3447
- Lijmer JG, Mol BW, Heisterkamp S, et al. Empirical evidence of design-related bias in studies of diagnostic tests. *JAMA*. 1999;282(11):1061-1066. doi:10.1001/jama.282.11.1061
- Klaiman C, White S, Richardson S, et al. Expert clinician certainty in diagnosing autism spectrum disorder in 16-30-month-olds. *J Autism Dev Disord*. Published November 17, 2022. doi:10.1007/s10803-022-05812-8
- McDonnell CGG, Bradley CCC, Kanne SMM, Lajonchere C, Warren Z, Carpenter LAA. When are we sure? predictors of clinician certainty in the diagnosis of autism spectrum disorder. *J Autism Dev Disord*. 2019;49(4):1391-1401. doi:10.1007/s10803-018-3831-3
- Lord C, Rutter M, DiLavore P, Risi S, Gotham K, Bishop S. *Autism Diagnostic Observation Schedule*. 2nd ed. Western Psychological Services; 2012.
- Mullen EM. *Mullen Scales of Early Learning*. American Guidance Service; 1995.
- Chawarska K, Klin A, Volkmar FR. In: Chawarska K, Klin A, Volkmar FR, eds. *Autism Spectrum Disorders in Infants and Toddlers*. Guilford Press; 2008.
- Agresti A, Coull B. Approximate is better than "exact" for interval estimation of binomial proportions. *Am Stat*. 1998;52(2):119-126.
- Bollen KA, Jackman RW. Regression diagnostics. *Sociol Methods Res*. 1985;13(4):510-542. doi:10.1177/0049124185013004004
- Kutner M, Nachtsheim C, Neter J, Li W. *Applied Statistical Linear Models*. 5th ed. McGraw-Hill; 2005.
- Linnet K. Performance of Deming regression analysis in case of misspecified analytical error ratio in method comparison studies. *Clin Chem*. 1998;44(5):1024-1031. doi:10.1093/clinchem/44.5.1024
- Martin RF. General deming regression for estimating systematic bias and its confidence interval in method-comparison studies. *Clin Chem*. 2000;46(1):100-104. doi:10.1093/clinchem/46.1.100
- Streiner DL, Norman GR. Validity: unreliability of the criterion. In: *Health Measurement Scales: A Practical Guide to Their Development and Use*. 4th ed. Oxford University Press; 2008:271-273.
- Law MR, Wald NJ, Wu T, Hackshaw A, Bailey A. Systematic underestimation of association between serum cholesterol concentration and ischaemic heart disease in observational studies: data from the BUPA study. *BMJ*. 1994;308(6925):363-366. doi:10.1136/bmj.308.6925.363
- Charles EP. The correction for attenuation due to measurement error. *Psychol Methods*. 2005;10(2):206-226. doi:10.1037/1082-989X.10.2.206
- Frankenburg WK, Dodds J, Archer P, Shapiro H, Bresnick B. The Denver II: a major revision and restandardization of the Denver Developmental Screening Test. *Pediatrics*. 1992;89(1):91-97. doi:10.1542/peds.89.1.91
- Altman DG, Bland JM. Diagnostic tests 2: predictive values. *BMJ*. 1994;309(6947):102. doi:10.1136/bmj.309.6947.102
- FDA-NIH Biomarker Working Group. *BEST (Biomarkers, EndpointS, and Other Tools) Resource*. Food and Drug Administration; National Institutes of Health; 2016.
- Ellis BJ, Boyce WT, Belsky J, Bakermans-Kranenburg MJ, van Ijzendoorn MH. Differential susceptibility to the environment. *Dev Psychopathol*. 2011;23(1):7-28. doi:10.1017/S0954579410000611